

TONGYEONG, Korea, Republic of

WMO: 471620

Lat: 34.8456N

Lon: 128.4356E

Elev: 109

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.9	26.7	-1.1	5.2	30.9	2.3	6.2	31.8	18.1	34.7	16.4	36.4	7.0	0	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	87.8	77.3	85.7	76.8	83.8	76.3	79.7	84.5	78.8	83.1	77.8	82.2	7.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	147.4	82.4	77.4	143.1	81.7	76.5	138.8	81.0	43.4	84.6	42.4	83.3	41.5	82.1	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	14.5	12.5	DB	19.1	91.7	3.1	2.5	16.8	93.5	15.1	95.0	13.3	96.4	11.1	98.2	(4)
(5)			WB	15.1	80.3	2.7	1.6	13.2	81.4	11.7	82.3	10.2	83.2	8.2	84.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	58.8	37.9	40.9	47.6	56.0	63.9	69.9	76.5	79.4	72.9	64.2	52.9	42.1	(6)
(7)		DBStd	14.72	5.48	5.97	5.46	4.61	3.61	2.75	3.95	3.49	3.87	4.86	6.30	6.21	(7)
(8)		HDD50	1048	375	259	110	8	0	0	0	0	0	0	43	253	(8)
(9)		HDD65	3532	839	675	540	270	63	1	0	0	0	72	363	709	(9)
(10)		CDD50	4260	1	4	36	189	432	597	822	912	688	440	130	9	(10)
(11)		CDD65	1271	0	0	0	1	30	149	357	447	239	47	1	0	(11)
(12)		CDH74	7883	0	0	0	1	63	274	2354	4013	1099	79	0	0	(12)
(13)		CDH80	1689	0	0	0	0	2	9	459	1080	138	1	0	0	(13)
(14)	Wind	WSAvg	5.5	5.9	6.0	6.2	5.8	5.2	4.8	5.6	5.5	5.3	5.2	5.2	5.7	(14)
(15)	Precipitation	PrecAvg	56.0	1.3	1.9	3.2	5.7	5.7	7.9	9.9	9.2	6.1	2.1	2.0	1.1	(15)
(16)		PrecMax	81.9	4.3	4.9	7.5	12.9	15.8	22.0	19.4	23.9	17.7	7.7	5.0	2.8	(16)
(17)		PrecMin	33.3	0.0	0.0	0.5	1.4	0.6	0.9	3.6	1.8	0.6	0.0	0.1	0.0	(17)
(18)		PrecStd	12.9	1.1	1.3	1.6	2.9	3.7	4.6	4.6	5.9	4.0	2.0	1.5	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.0	60.1	63.7	71.7	79.0	81.1	90.4	91.6	85.9	78.6	70.1	60.9	(19)
(20)			MCWB	50.6	53.9	53.4	57.1	62.7	67.9	77.8	79.1	75.1	66.7	61.0	53.6	(20)
(21)		2%	DB	51.7	56.2	61.1	68.3	75.5	78.6	86.8	88.9	83.4	76.2	67.2	57.3	(21)
(22)			MCWB	43.9	49.2	53.2	56.6	61.8	67.4	77.5	77.9	73.1	65.6	58.6	50.3	(22)
(23)		5%	DB	48.9	53.4	59.1	65.9	73.1	76.9	84.4	86.9	81.5	74.2	64.7	54.2	(23)
(24)			MCWB	41.6	45.9	51.4	56.6	61.3	67.8	77.0	77.1	72.4	64.6	56.9	46.9	(24)
(25)		10%	DB	46.4	50.6	56.9	63.9	70.8	75.5	82.4	85.2	79.8	72.1	62.5	51.7	(25)
(26)			MCWB	39.8	43.3	49.4	55.6	61.4	68.2	76.2	76.9	71.6	63.0	55.4	45.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	57.1	59.0	64.2	68.3	74.4	80.4	81.2	78.2	72.4	64.3	56.2	(27)
(28)			MCDWB	55.2	58.7	60.5	66.3	70.9	76.0	85.5	86.9	82.4	75.2	67.2	58.8	(28)
(29)		2%	WB	45.9	51.4	56.3	61.5	66.2	72.9	79.2	80.0	76.9	69.8	61.7	51.5	(29)
(30)			MCDWB	49.6	54.4	58.9	64.2	70.2	74.7	83.5	85.1	80.9	73.8	64.9	55.4	(30)
(31)		5%	WB	43.4	47.2	53.4	59.3	64.8	71.6	78.3	79.3	75.3	67.0	59.1	48.6	(31)
(32)			MCDWB	47.3	51.8	57.4	63.2	69.2	73.8	82.4	84.0	79.3	71.5	63.2	52.9	(32)
(33)		10%	WB	40.9	44.2	50.6	57.5	63.5	70.1	77.3	78.5	73.5	64.7	56.8	45.9	(33)
(34)			MCDWB	45.6	49.1	55.4	61.7	68.2	73.3	81.3	83.0	77.6	70.0	61.3	50.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.3	13.6	13.4	13.0	12.0	9.8	8.2	9.3	11.0	13.3	13.7	13.7	(35)
(36)			MCDBR	14.8	15.7	14.8	15.0	15.5	12.8	10.9	11.7	12.7	14.2	14.5	15.1	(36)
(37)			MCWBR	10.1	10.7	9.4	7.9	6.9	5.7	4.1	4.2	5.6	6.6	8.5	10.4	(37)
(38)		5% WB	MCDWB	12.4	13.8	12.4	10.9	10.9	7.4	9.3	9.2	9.6	10.9	12.1	13.4	(38)
(39)			MCWBR	10.4	11.6	11.0	8.8	6.9	5.1	4.3	4.1	5.6	7.1	9.0	10.8	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.414	0.463	0.535	0.562	0.576	0.617	0.589	0.546	0.497	0.459	0.429	0.407	(41)	
(42)		taud	2.073	1.979	1.822	1.810	1.831	1.814	1.942	2.058	2.137	2.118	2.115	2.105	(42)	
(43)		Ebn at Noon	239	241	234	236	235	225	230	238	243	240	233	233	(43)	
		Edh at Noon	41	50	63	67	67	68	59	52	46	43	39	37		
(44)	All-Sky Solar Radiation	RadAvg	858	1071	1394	1600	1754	1545	1487	1512	1284	1180	888	783	(44)	
(45)		RadStd	68	108	118	125	194	137	219	185	133	108	98	43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	N/A

Nomenclature: See separate page